Arithmetic and Order of Real Numbers

- **Definition 1**
  The set of **real numbers**, denoted $\mathbb{R}$, is a set that contains $\mathbb{Q}$. Its sum, product, and order extend the operations and order in $\mathbb{Z}$ and satisfy the following properties for all $x, y, z \in \mathbb{R}$:

  1. $$(x+y)+z = x+(y+z).$$
  2. There exists a unique element $0 \in \mathbb{R}$ such that
  $$x+0 = x.$$
  3. For each $x \in \mathbb{R}$, there exists a unique $-x \in \mathbb{R}$ such that
  $$x+(-x) = 0.$$
  4. $$x+y = y+x.$$
  5. $$(xy)z = x(yz).$$
  6. There exists a unique element $1 \in \mathbb{R}$ such that
  $$x \cdot 1 = x.$$
  7. For each $x \in \mathbb{R}$ with $x \neq 0$, there exists a unique element $x^{-1} \in \mathbb{R}$ such that
  $$x \cdot x^{-1} = 1.$$
  8. $$xy = yx.$$
  9. $$x(y+z) = xy+xz.$$
  10. The relation $\leqslant$ is a total order in $\mathbb{R}$.
  11. If $x \leqslant y$, then $x+z \leqslant y+z$.
  12. If $x \leqslant y \wedge z \geqslant 0$, then $xz \leqslant yz$.

- **Definition 2**
  The set of **irrational numbers** is the set of those real numbers that are not rational numbers, that is,
  $$\mathbb{R} \setminus \mathbb{Q}.$$

  - **Example 2.1**
  $$\sqrt{2}, \mathrm{e}, \pi.$$

  - **Proposition 2.2**
    The number $\sqrt{2}$ is irrational.

    - **Proof 2.2.1**
      Suppose that $\sqrt{2} = \frac{p}{q}$ with $p \in \mathbb{Z}, q \in \mathbb{N}$ and p, q are coprime. Then
      $$p^2 = 2q^2.$$
      Hence $2 \mid p^2$, and since 2 is a prime we have that $2 \mid p$, that is, there exists $k \in \mathbb{Z}$ such that $p = 2k$. So

$$\begin{aligned}4k^2 &= 2q^2,\\ 2k^2 &= q^2.\end{aligned}$$

Then $2 \mid q^2$, and since 2 is a prime we have that $2 \mid q$, which is a contradiction since p, q are coprime.

- **Proposition 2.3**
  Let $a, b, c, d \in \mathbb{R}$, then:
  1. If $a < b$ and $c < 0$, then $ac > bc$.
  2. If $a < b$ and $c < d$, then $a + c < b + d$.
  3. If $a < b$, then $-a > -b$.
  4. If $a < 0$ and $b < 0$, then $ab > 0$.

  - **Proof 2.3.1**
    1. Since $-c > 0$ and $b - a > 0$, then
    $$ac - bc = (b - a)(-c) > 0.$$
    And thus $ac > bc$.
    2. Since $b - a > 0$ and $d - c > 0$, then
    $$(b + d) - (a + c) = (b - a) + (d - c) > 0.$$
    And thus $a + c < b + d$.
    3. By $\text{Proposition } 2.3(1)$ with $c = -1$.
    4. Since $-a > 0$ and $-b > 0$, then $ab = (-a)(-b) > 0$.

- **Definition 3**
  Let $x \in \mathbb{R}$ and $n \in \mathbb{N}$, we define
  $$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ times}}.$$
  If $x \neq 0$ we define $x^0 = 1$ and $x^{-n} = \left(x^{-1}\right)^n$.

  - **Theorem 3.1**
    Let $x, y \in \mathbb{R}$ and $m, n \in \mathbb{N}$, then
    1. $x^{m+n} = x^m \cdot x^n$.
    2. $(x^m)^n = x^{mn}$.
    3. $(xy)^n = x^n y^n$.
    4. $\dfrac{x^m}{x^n} = x^{m-n}$ if $x \neq 0$.
    5. $\left(\dfrac{x}{y}\right)^n = \dfrac{x^n}{y^n}$ if $y \neq 0$.

    - **Proof 3.1.1**
      1. Fix $m \in \mathbb{N}$ and use induction on n.
      For $n = 1$ the result is true since $x^{m+1} = x^m x = x^m x^1$.
      Suppose that the result is true for n, that is
      $$x^{m+n} = x^m x^n.$$
      Hence

$$
\begin{aligned}
x^{m+(n+1)} &= x^{(m+n)+1} \\
&= x^{m+n} x \\
&= \left(x^m x^n\right) x \\
&= \left(x^m\right)\left(x^n x\right) \\
&= x^m x^{n+1}.
\end{aligned}
$$

Thus the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

2. Fix $m \in \mathbb{N}$ and use induction on n.
For $n=1 the result is true since $\left(x^m\right)^1 = x^m = x^{m \cdot 1}$.
Suppose that the result is true for n, that is

$$\left(x^m\right)^n = x^{mn}.$$

Hence

$$
\begin{aligned}
\left(x^m\right)^{n+1} &= \left(x^m\right)^n \cdot x^m \\
&= x^{mn} \cdot x^m \\
&= x^{mn+m} \\
&= x^{m(n+1)}.
\end{aligned}
$$

Thus the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

3. Induction on n.
For $n=1$ the result is true since $(xy)^1 = xy = x^1 y^1$.
Suppose that the result is true for n, that is

$$(xy)^n = x^n y^n.$$

Hence

$$
\begin{aligned}
(xy)^{n+1} &= (xy)^n (xy) \\
&= x^n y^n (xy) \\
&= \left(x^n x\right)\left(y^n y\right) \\
&= x^{n+1} y^{n+1}.
\end{aligned}
$$

Therefore the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

4. Fix $m \in \mathbb{N}$ and suppose $x \neq 0$. Use induction on n.
For $n=1$ the result is true since $\dfrac{x^m}{x^1} = x^m x^{-1} = x^{m+(-1)} = x^{m-1}$ by Property 1.
Suppose that the result is true for n, that is

$$\frac{x^m}{x^n} = x^{m-n}.$$

Hence

$$
\begin{aligned}
\frac{x^m}{x^{n+1}} &= \frac{x^m}{x^n \cdot x} \\
&= \frac{x^m}{x^n} \cdot x^{-1} \\
&= x^{m-n} \cdot x^{-1} \\
&= x^{(m-n)+(-1)} \\
&= x^{m-(n+1)}.
\end{aligned}
$$

Thus the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

5. Suppose $y \neq 0$. Use induction on n.
   For $n = 1$ the result is true since $\left(\frac{x}{y}\right)^1 = \frac{x}{y} = \frac{x^1}{y^1}$.
   Suppose that the result is true for n, that is

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}.$$

   Hence

$$\begin{aligned}\left(\frac{x}{y}\right)^{n+1} &= \left(\frac{x}{y}\right)^n \cdot \left(\frac{x}{y}\right) \\ &= \frac{x^n}{y^n} \cdot \left(\frac{x}{y}\right) \\ &= \frac{x^n \cdot x}{y^n \cdot y} \\ &= \frac{x^{n+1}}{y^{n+1}}.\end{aligned}$$

   Thus the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

- **Proposition 3.2**
  If $n \in \mathbb{N}$ and $0 \leqslant x < y$, then $x^n < y^n$. In particular, if $x, y \geqslant 0$ and $x^n = y^n$, then $x = y$.
  - **Proof 3.2.1**
    Use induction on n.
    If $n = 1$, then the result is true since $x^1 < y^1$.
    Suppose that the result is true for n, that is,

$$x^n < y^n.$$

    Hence,

$$x^{n+1} = x^n x \leqslant x^n y < y^n y = y^{n+1}.$$

    Thus the result is true for $n+1$. So the result is true for every $n \in \mathbb{N}$.

    Suppose that $x, y \geqslant 0$, $x^n = y^n$ and $x \neq y$.
    If $x < y$, then $x^n < y^n$, which is a contradiction.
    If $x > y$, then $x^n > y^n$, which is a contradiction.
    Therefore $x = y$.

- **Definition 4**
  If $x \geqslant 0$ and $n \in \mathbb{N}$, the **n-th root** of x, denoted $\sqrt[n]{x}$, is the unique nonnegative number y such that $y^n = x$.
  - **Example 4.1**
    The equation $y^2 = 9$ has two solutions: -3 and 3. But $\sqrt{9} = 3$ denotes only the nonnegative one.
  - **Remark 4.2**
    A negative real number has no even root, whereas an odd root is defined by

$$\sqrt[n]{x} = -\sqrt[n]{-x}$$

    if $x < 0$ and n is odd.

  If $x^2 < y^2$, must $x < y$? No. For example, $(-1)^2 < (-2)^2$ but $-1 > -2$. The implication is true when both numbers are nonnegative.

  If $x, y \geqslant 0$ with $x^n < y^n$, then $x < y$. By $\text{Proposition } 3.2$, since if $x = y$, then $x^n = y^n$, and if $x > y$, then $x^n > y^n$, which is a contradiction.